

$$b) x^2 - 4x + 6 = 0$$

$$x = \frac{4 \pm 2i}{2} \Rightarrow \boxed{2 \pm i}$$

③ $f(x) = z^2 - z + 1 \rightarrow f(1-i) = (1-i)^2 - (1-i) + 1 \rightarrow 1 - 2i + i^2 - 1 + i + 1 \rightarrow -i$

5) a) parte real: $x=7$ parte imaginario: $y=3$ b) $3x + 4ix + 3yi + 4yi^2 = 7 + 26i \rightarrow \begin{cases} 3x - 4y = 7 \rightarrow x = \frac{7+4y}{3} \\ (4x+3y)i = 26i \rightarrow 4x+3y=26 \end{cases}$

* Ao final das relações temos $y = 2$, $x = 5$

6) a) $\frac{2}{i} - \frac{i}{i} \rightarrow \frac{-2i}{i^2} \rightarrow 2i$ b) $\frac{3}{2+i} \cdot \frac{(2-i)}{(2-i)} \rightarrow \frac{6-3i}{4+1} \} \frac{6-3i}{5}$ c) $\frac{i^9}{4-3i} \} \frac{i}{4-3i} \cdot \frac{(4+3i)}{(4+3i)} \rightarrow \frac{4i-3}{25}$ d) $\frac{7+i}{7-i} \cdot \frac{(7+i)}{(7+i)} \rightarrow \frac{7+i+i^2}{2}$

$$\frac{2i}{2} \rightarrow (i)$$

① $f = (x^2 + \sqrt{2}x + 1) \cdot (x^2 - \sqrt{2}x + 1) \rightarrow (a+b) \cdot (a-b) \rightarrow a^2 - ab + ab - b^2 \rightarrow a^2 - b^2$

$$\underbrace{(x^2+1)}_a \cdot \underbrace{(x^2+1)}_a \cdot \underbrace{(1+\sqrt{2}x)}_b \cdot \underbrace{(1-\sqrt{2}x)}_b \rightarrow a^2 - b^2 \rightarrow (x^2+1)^2 - (\sqrt{2}x)^2 \rightarrow x^4 + 1 = 8$$

(2)
$$\begin{array}{r|l} 3x^3 - x^4 + 2x^3 + 0x^2 + 4x - 3 & x^3 + 0x^2 - 2x + 7 \\ \hline -3x^3 + 0x^2 + 6x^3 + 3x^2 & \\ \hline -x^4 + 8x^3 + 3x^2 + 4x & \\ -x^4 + 0x^3 - 2x^2 + x & \\ \hline 8x^3 + x^2 + 5x - 3 & \\ -8x^3 + 0x^2 + 16x - 8 & \\ \hline x^2 + 21x - 11 & \end{array}$$

$$\begin{array}{r|l} x^4 + 0x^3 + 0x^2 - 2x + 13 & x^3 + x + 1 \\ -x^4 + 1x^3 + x^2 & \\ \hline x^3 + x^2 - 2x & \\ -x^3 - x^2 - x & \\ \hline -x + 13 & \end{array}$$

$$\begin{array}{r|l} 2x^5 + 0x^4 + 0x^3 + 0x^2 - 3x + 7 & x^2 + 0x + 1 \\ -2x^5 - 0x^4 - 2x^3 & \\ \hline & 2x^3 - 3x + 7 \\ & -2x^3 + 0x^2 - 3x & \\ \hline & 0x^3 + 0x^2 - 6x + 7 & \\ & +2x^3 - 0x^2 + 2x & \\ \hline & -4x^3 + 2x + 7 & \end{array}$$

③

3	2	-5	0	-70	-7
	2	1	3	-7	-4

$\hookrightarrow$ Presto

$$2x^3 + x^2 + 3x - 7$$

$$\begin{array}{c|ccc} \textcircled{4} & -1 & 1 & 1 & 1 \\ \hline & & 1 & 0 & \textcircled{1} \end{array}$$

* Poderia ter usado o teorema da reta.

O resto do diviso é 7,

⑤ *A raiz é pelo termo de resto $\rightarrow x-3=0 \rightarrow x=3$

$$0(9)^2 - 2(3) + 7 = 4 \rightarrow 90 - 6 + 7 = 4 \rightarrow 90 - 5 = 4 \rightarrow 90 : 9 \rightarrow 0 = 10$$

$$\begin{array}{r|l} \textcircled{6} & x^2 + 0x - 2 \\ -2x^3 - 3x^2 + 0x + 6 & \\ \hline -2x^3 + 0x^2 + 4x & \\ \hline -3x^2 + 4x & 6 \\ +3x^2 + 0x & -6 \\ \hline 4x + 0 & \\ \hline 4x & \end{array}$$

$$\begin{array}{r|l} b) \quad 6x^2 - 4x - 3 & 3x - 5 \\ \underline{-6x^2 + 10x} & 2x + 2 \\ & \underline{6x - 3} \\ & \underline{-6x + 10} \\ & 7 \end{array}$$

$$\begin{array}{r|l} 3x+7 & x+4 \\ -3x-7 & 3 \\ \hline & -5 \end{array}$$

$$\begin{array}{r|l} 2x^3 + 0x^2 - 4x - 1 & 2x - 1 \\ -2x^3 + 72x^2 & \\ \hline & 72x^2 + 6x - 1 \\ & \vdots \\ & 22x^2 - 4x \\ & -22x^2 + 6x \\ \hline & 2x - 1 \\ & -2x + 1 \\ \hline & 0 \end{array}$$

$$\begin{array}{r|l} 2) & 6x^2 + 7x + 15 \\ & -6x^2 + 15x - 3 \\ \hline & 0 \quad 22x + 12 \\ & \underline{22x + 12} \\ & 0 \end{array}$$

$$22x + 72 =$$

7) a) $-2 \mid 1 \ 0 \ 0 \ 2 \ -72$
 $\begin{array}{r} 1 \\ -2 \\ \hline 1 \end{array}$ $\begin{array}{r} 0 \\ -2 \\ \hline -2 \end{array}$ $\begin{array}{r} 0 \\ 4 \\ \hline 4 \end{array}$ $\begin{array}{r} 2 \\ -72 \\ \hline -74 \end{array}$ $\textcircled{0}$ $\hookrightarrow$ Resto
 $x^3 - 2x^2 + 4x - 6$

b) $2 \mid 3 \ 2 \ -5$
 $\begin{array}{r} 3 \\ 6 \\ \hline -4 \end{array}$ $\begin{array}{r} 2 \\ 4 \\ \hline -2 \end{array}$ $\begin{array}{r} -5 \\ 10 \\ \hline 5 \end{array}$ $\textcircled{17}$ $\hookrightarrow$ Resto
 $3x + 8$

c) $3 \mid 1 \ 0 \ -9 \ +0 \ +2 \ +0$
 $\begin{array}{r} 1 \\ 3 \\ \hline -2 \end{array}$ $\begin{array}{r} 0 \\ 3 \\ \hline -3 \end{array}$ $\begin{array}{r} -9 \\ 27 \\ \hline 18 \end{array}$ $\begin{array}{r} 0 \\ 6 \\ \hline -6 \end{array}$ $\begin{array}{r} 2 \\ 6 \\ \hline -4 \end{array}$ $\begin{array}{r} 0 \\ -12 \\ \hline 12 \end{array}$ $\begin{array}{r} 0 \\ 36 \\ \hline 36 \end{array}$ $\textcircled{6}$ $\hookrightarrow$ Resto
 $x^4 + 3x^3 + 2$

d) $-\frac{1}{2} \mid 6 \ -7 \ -9$
 $\begin{array}{r} 6 \\ -3 \\ \hline 9 \end{array}$ $\begin{array}{r} -7 \\ 3.5 \\ \hline -3.5 \end{array}$ $\begin{array}{r} -9 \\ 4.5 \\ \hline -4.5 \end{array}$ $\textcircled{-4}$ $\hookrightarrow$ Resto
 $6x - 70$

e) $4 \mid -4 \ 11 \ 26$
 $\begin{array}{r} -4 \\ 16 \\ \hline 12 \end{array}$ $\begin{array}{r} 11 \\ 44 \\ \hline -33 \end{array}$ $\begin{array}{r} 26 \\ 164 \\ \hline -138 \end{array}$ $\textcircled{6}$ $\hookrightarrow$ Resto
 $-4x - 5$

f) $2 \mid 5 \ 0 \ 0 \ 0 \ -7$
 $\begin{array}{r} 5 \\ 10 \\ \hline -5 \end{array}$ $\begin{array}{r} 0 \\ 10 \\ \hline -10 \end{array}$ $\begin{array}{r} 0 \\ 20 \\ \hline -20 \end{array}$ $\begin{array}{r} 0 \\ 40 \\ \hline -40 \end{array}$ $\begin{array}{r} -7 \\ 14 \\ \hline 7 \end{array}$ $\textcircled{79}$ $\hookrightarrow$ Resto
 $x^3 + 70x^2 + 20x + 70$

g) $3 \mid 1 \ 0 \ 0 \ 0 \ 0$
 $\begin{array}{r} 1 \\ 3 \\ \hline -2 \end{array}$ $\begin{array}{r} 0 \\ 6 \\ \hline -6 \end{array}$ $\begin{array}{r} 0 \\ 18 \\ \hline -18 \end{array}$ $\begin{array}{r} 0 \\ 27 \\ \hline -27 \end{array}$ $\begin{array}{r} 0 \\ 81 \\ \hline -81 \end{array}$ $\textcircled{81}$ $\hookrightarrow$ Resto
 $x^3 + 3x^2 + 9x + 27$

8) * Vamos dividir o termo da fatura utilizando o termo da soma:

a) $x - 8 = 0 \rightarrow x = 8 \quad b^2 - 9 \cdot 8 + c = 0 \rightarrow 64 - 72 + c = 0 \rightarrow c = 8$

b) $x + 3 \rightarrow x = -3 \quad 5 \cdot (-3)^2 - 3c + 9 = 0 \rightarrow -3c = -54 \rightarrow c = 18$

c) $x - 5 \rightarrow x = 5 \quad 125 - 750 + 75 + c = 0 \rightarrow c = 70$

d) $c = 7 \quad e) c = -75 \quad f) c = 0$

9) Pela soma da restos:

a) $2 - 8 \cdot 4^3 + 4 \cdot 2^2 + 75 \cdot 2 + 6 \rightarrow \boxed{x = 4}$ b) $x + 1 = 0 \rightarrow x = -1 \quad 1 - 7^2 - 7 + 7 = \boxed{x = 7}$

c) $x + 2k = 0 \rightarrow x = -2k \quad f(-2k) = k \cdot (-2k)^2 - 2k - 7 \rightarrow \boxed{x = 4k^2 - 2k - 7}$

10) a) $x^3 - 4x \rightarrow x \cdot (x^2 - 4) \rightarrow x \cdot (x - 2) \cdot (x + 2)$ b) $x^3 - 4x^2 - 21x = 0 \rightarrow x \cdot (x^2 - 4x - 21) = 0 \rightarrow x \cdot (x - 7) \cdot (x + 3) = 0$

c) $2x^3 + 7x^2 - 6x \rightarrow x \cdot (2x^2 + 7x - 6) \rightarrow x \cdot (x - 6) \cdot (2x - 1)$ d) $-3x^3 + 6x^2 + 9x \rightarrow 3x \cdot (-x^2 + 2x + 3) \rightarrow 3x \cdot (x - 3) \cdot (x + 1)$

e) $x^4 - x^3 - 20x^2 \rightarrow x^2 \cdot (x^2 - x - 20) \rightarrow x^2 \cdot (x - 5) \cdot (x + 4)$ f) $8x^4 - 6x^3 - 2x^2 \rightarrow 2x^2 \cdot (4x^2 - 3x - 1) \rightarrow 2x^2 \cdot (x - 1) \cdot (4x + 1)$

11) a) * Sabemos que $x = -1$ é raiz, portanto, podemos concluir que $x + 1$ consegue dividir o polinômio e dar os outros raízes.
 $\hookrightarrow$ Podemos aplicar o lagrange para os outros quocientes do mesmo forma.

$\begin{array}{r} x^3 + x^2 - 2x - 2 \\ -x^3 - x^2 \\ \hline 0 - 2x - 2 \\ +2x + 2 \\ \hline 0 \end{array}$ $\textcircled{1}$ $\hookrightarrow$ As outras duas raízes são $x = \pm i\sqrt{2}$ e $x = -1$

b) $\begin{array}{r} x^3 + x^2 - 4x - 20 \\ -x^3 + 2x^2 \\ \hline 3x^2 - 4x \\ -3x^2 + 6x \\ \hline -10x - 20 \\ +10x + 20 \\ \hline 0 \end{array}$ $\textcircled{2}$ $\textcircled{3}$ Raízes: $(-5; -2; 2)$

c) $4x^4 - 21x^3 - 19x^2 + 6x \rightarrow x \cdot (4x^3 - 21x^2 - 19x + 6) \rightarrow x \cdot (4x + 1) \cdot (x^2 - 5x + 6)$

$\begin{array}{r} 4x^3 - 21x^2 - 19x + 6 \\ -4x^3 + 4x^2 \\ \hline -17x^2 - 19x + 6 \\ +17x^2 - 85x + 102 \\ \hline -104x + 108 \\ +104x - 104 \\ \hline 4 \end{array}$ $\textcircled{4}$ $\textcircled{5}$ $\textcircled{6}$ $\textcircled{7}$ Raízes: $(0; -\frac{1}{4}; 6; -7)$

* Como temos raiz dupla, podemos usar o método dos restos.

d) $x^4 + 2x^3 - 5x^2 - 36x + 60$

$\downarrow$
 $(x - 2)^2 \cdot (x^2 + 6x + 15)$ $\textcircled{7}$ Raízes: $(2; 2; -3 + i\sqrt{6}; -3 - i\sqrt{6})$

* Resultado das operações com Restos:

e) $6x^4 + 7x^3 + 6x^2 - 7 = 0$ é dividível por $(3x - 1)$ e $(2x + 1) \rightarrow 6x^2 + x - 7$

$\begin{array}{r} 6x^4 + 7x^3 + 6x^2 - 7 \\ 6x^4 + 4x^3 + 2x^2 - 7 \\ \hline 3x^3 + 4x^2 - 7 \end{array}$ $\textcircled{8}$ $\textcircled{9}$ $\textcircled{10}$ $\textcircled{11}$ $\textcircled{12}$ $\textcircled{13}$ $\textcircled{14}$ $\textcircled{15}$ $\textcircled{16}$ $\textcircled{17}$ $\textcircled{18}$ $\textcircled{19}$ $\textcircled{20}$ $\textcircled{21}$ $\textcircled{22}$ $\textcircled{23}$ $\textcircled{24}$ $\textcircled{25}$ $\textcircled{26}$ $\textcircled{27}$ $\textcircled{28}$ $\textcircled{29}$ $\textcircled{30}$ $\textcircled{31}$ $\textcircled{32}$ $\textcircled{33}$ $\textcircled{34}$ $\textcircled{35}$ $\textcircled{36}$ $\textcircled{37}$ $\textcircled{38}$ $\textcircled{39}$ $\textcircled{40}$ $\textcircled{41}$ $\textcircled{42}$ $\textcircled{43}$ $\textcircled{44}$ $\textcircled{45}$ $\textcircled{46}$ $\textcircled{47}$ $\textcircled{48}$ $\textcircled{49}$ $\textcircled{50}$ $\textcircled{51}$ $\textcircled{52}$ $\textcircled{53}$ $\textcircled{54}$ 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